then

$$d\omega = \frac{dx}{y^2}, \quad = \frac{dx}{X^{\frac{2}{3}}};$$

and corresponding to the differential relation

$$X_1^{-\frac{1}{3}}\,dx_1 + X_2^{-\frac{1}{3}}\,dx_2 + X_3^{-\frac{1}{3}}\,dx_3 = 0,$$

we have the integral relation

$$\nabla, = \begin{vmatrix} X_1^{\frac{1}{3}}, & X_2^{\frac{1}{3}}, & X_3^{\frac{1}{3}} \\ x_1, & x_2, & x_3 \\ 1, & 1, & 1 \end{vmatrix} = 0,$$

viz. this last equation, as containing no arbitrary constant, is a particular integral of the differential equation.

If instead of x_1, x_2, x_3 we write x, y, z, then we have

$$\nabla \;= \begin{vmatrix} X^{\frac{1}{3}}, & Y^{\frac{1}{3}}, & Z^{\frac{1}{3}} \\ x, & y, & z \\ 1, & 1, & 1 \end{vmatrix}, = 0,$$

as a particular integral of the differential equation

$$X^{-\frac{1}{3}}\,dx + Y^{-\frac{1}{3}}\,dy + Z^{-\frac{1}{3}}\,dz = 0.$$

To rationalize the integral equation, write

$$\alpha,\ \beta,\ \gamma = y - z,\ z - x,\ x - y \quad (\text{so that } \alpha + \beta + \gamma = 0),$$

the equation is

$$\alpha X^{\frac{1}{3}} + \beta Y^{\frac{1}{3}} + \gamma Z^{\frac{1}{3}} = 0;$$

and we thence have

$$\alpha^3 X + \beta^3 Y + \gamma^3 Z = 3\alpha\beta\gamma\, X^{\frac{1}{3}} Y^{\frac{1}{3}} Z^{\frac{1}{3}}.$$

The left-hand side is

$$\begin{aligned} &\alpha^3\,(A + 3Bx + 3Cx^2 + Dx^3) \\ &\quad + \beta^3\,(A + 3By + 3Cy^2 + Dy^3) \\ &\quad + \gamma^3\,(A + 3Bz + 3Cz^2 + Dz^3); \end{aligned}$$

or assuming

$$\alpha',\ \beta',\ \gamma' = x(y - z),\ y(z - x),\ z(x - y) \quad (\text{so that } \alpha' + \beta' + \gamma' = 0),$$

this is

$$= A(\alpha^3 + \beta^3 + \gamma^3) + 3B(\alpha^2\alpha' + \beta^2\beta' + \gamma^2\gamma') + 3C(\alpha\alpha'^2 + \beta\beta'^2 + \gamma\gamma'^2) + D(\alpha'^3 + \beta'^3 + \gamma'^3).$$

But taking λ arbitrary, and

$$a,\ b,\ c = \alpha + \lambda\alpha',\ \beta + \lambda\beta',\ \gamma + \lambda\gamma',$$

then

$$a + b + c = 0,$$

whence

$$a^3 + b^3 + c^3 = 3abc;$$

or substituting for a, b, c their values, and comparing the coefficients of the several powers of λ,

$$\begin{aligned}
\alpha^3+\beta^3+\gamma^3 &= 3\alpha\beta\gamma,\\
\alpha^2\alpha'+\beta^2\beta'+\gamma^2\gamma' &= \alpha^2\beta'\gamma'+\beta^2\gamma'\alpha'+\gamma^2\alpha'\beta' = \alpha\beta\gamma\,(x+y+z),\\
\alpha\alpha'^2+\beta\beta'^2+\gamma\gamma'^2 &= \alpha\alpha'^2\beta\gamma+\beta\beta'^2\gamma\alpha+\gamma\gamma'^2\alpha\beta = \alpha\beta\gamma\,(yz+zx+xy),\\
\alpha'^3+\beta'^3+\gamma'^3 &= 3\alpha'\beta'\gamma' = 3\alpha\beta\gamma\cdot xyz.
\end{aligned}$$

Hence we have

$$\alpha^3X+\beta^3Y+\gamma^3Z = 3\alpha\beta\gamma\,[A+B(x+y+z)+C(yz+zx+xy)+Dxyz],$$

or the integral equation is

$$[A+B(x+y+z)+C(yz+zx+xy)+Dxyz]^3 = XYZ,$$

that is,

$$\sqrt[3]{\{A+B(x+y+z)+C(yz+zx+xy)+Dxyz\}^3} = \sqrt[3]{XYZ},$$

the elegant result given by Capt. MacMahon at the beginning of his paper.

The author in a letter to me, dated Jan. 13, 1883, remarks that the particular integral of the equation in question

$$X^{-\frac{1}{3}}\,dx+Y^{-\frac{1}{3}}\,dy+Z^{-\frac{1}{3}}\,dz=0,$$

is expressible as a determinant in a rational form as follows. Writing it $XYZ=P^3$, where

$$P = A+B(x+y+z)+C(yz+zx+xy)+Dxyz,$$

then the form is

$$\nabla,\ = \begin{vmatrix} 1, & \left(\frac{1}{3}P\dfrac{dX}{dx}-X\dfrac{dP}{dx}\right), & X \\ 1, & \left(\frac{1}{3}P\dfrac{dY}{dy}-Y\dfrac{dP}{dy}\right), & Y \\ 1, & \left(\frac{1}{3}P\dfrac{dZ}{dz}-Z\dfrac{dP}{dz}\right), & Z \end{vmatrix} = 0,$$

for, as shown by Captain MacMahon in his paper, each of the three terms such as

$$Z\left(\tfrac{1}{3}P\frac{dY}{dy}-Y\frac{dP}{dy}\right)-Y\left(\tfrac{1}{3}P\frac{dZ}{dz}-Z\frac{dP}{dz}\right),$$

which compose the determinant, is divisible by $XYZ-P^3$.

It may be added that we have identically

$$\tfrac{1}{3}P\frac{dX}{dx}-X\frac{dP}{dx} = (AC-B^2)(2x-y-z)+(AD-BC)(x^2-yz)+(BD-C^2)(x^2y+x^2z-2xyz),$$

and of course like values for the other two expressions in the determinant.

803.

ON MR ANGLIN'S FORMULA FOR THE SUCCESSIVE POWERS OF THE ROOT OF AN ALGEBRAICAL EQUATION.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xix. (1883), pp. 223, 224.]

Suppose $x^m - px^{m-1} + qx^{m-2} - \ldots = 0$, then the successive powers x^m, x^{m+1}, x^{m+2}, &c. of x can be expressed in the form $Px^{m-1} - Qx^{m-2} + Rx^{m-3} -$ &c. Mr Anglin has obtained for this purpose a very elegant formula, with a demonstration which (it occurred to me) might be presented under a somewhat simplified form; and he has permitted me to draw up the present Note.

Take, for greater convenience, the equation to be

$$x^4 - px^3 + qx^2 - rx + s = 0,$$

and let h_1, h_2, h_3, ... be the sums of the homogeneous products of the roots, of the orders 1, 2, 3, &c. respectively; then, writing also $h_0 = 1$, we have

$$\begin{aligned}
h_1 &= h_0 p, \\
h_2 &= h_1 p - h_0 q, \\
h_3 &= h_2 p - h_1 q + h_0 r, \\
h_4 &= h_3 p - h_2 q + h_1 r - h_0 s, \\
h_5 &= h_4 p - h_3 q + h_2 r - h_1 s, \\
&\ldots
\end{aligned}$$

And this being so, starting from the equation

$$x^4 = px^3 - qx^2 + rx - s,$$

that is,

$$= h_1 x^3 - h_0 q x^2 + h_0 r x - h_0 s,$$

we obtain successively

$$\begin{aligned}
x^5 &= h_1(px^3 - qx^2 + rx - s) \\
&\quad - h_0qx^3 + h_0rx^2 - h_0sx \\
&= h_2x^3 - (h_1q - h_0r)x^2 + (h_1r - h_0s)x - h_1s, \\
x^6 &= h_2(px^3 - qx^2 + rx - s) \\
&\quad - (h_1q - h_0r)x^3 + (h_1r - h_0s)x^2 - h_1sx \\
&= h_3x^3 - (h_2q - h_1r + h_0s)x^2 + (h_2r - h_1s)x - h_2s, \\
x^7 &= h_3(px^3 - qx^2 + rx - s) \\
&\quad - (h_2q - h_1r + h_0s)x^3 + (h_2r - h_1s)x^2 - h_2sx \\
&= h_4x^3 - (h_3q - h_2r + h_1s)x^2 + (h_3r - h_2s)x - h_3s,
\end{aligned}$$

and so on, the characteristic feature being that by the introduction of the symbols h, the coefficient of x^3 presents itself at each step as a monomial, and the coefficients of the lower powers require no reduction. It is obvious that the process is a perfectly general one, and that for the equation

$$x^m - p_1x^{m-1} + p_2x^{m-2} - \ldots + (-)^m p_m = 0,$$

the formula is

$$\begin{aligned}
x^{m+s} = \quad & h_{s+1}x^{m-1} \\
& - (h_sp_2 - h_{s-1}p_3 + \ldots)\, x^{m-2} \\
& + (h_sp_3 - h_{s-1}p_4 + \ldots)\, x^{m-3} \\
& \vdots \\
& + (-)^{m-1}(h_sp_{m-1} - h_{s-1}p_{m-2} + \ldots)\, x \\
& + (-)^{m-1}\, h_sp_m,
\end{aligned}$$

where, as regards each power of x, the series forming the coefficient thereof is continued as far as possible, that is, up to the term which contains p_m or h_0 as the case may be.

804.

ON THE ELLIPTIC-FUNCTION SOLUTION OF THE EQUATION $x^3 + y^3 - 1 = 0$.

[From the *Proceedings of the Cambridge Philosophical Society*, vol. iv. (1883), pp. 106–109.]

I HAD occasion to find elliptic-function expressions for the coordinates $(x,\ y)$ of a point on the cubic curve $x^3 + y^3 = 1$. These are derivable from the formulæ given, Legendre, *Fonctions Elliptiques*, t. I. pp. 185, 186, for the reduction to elliptic integrals of the integral $R = \int \frac{dz}{(1-z^3)^{\frac{1}{2}}}$. Legendre, writing

$$z = \frac{\sqrt{4y^3-1} - \sqrt{3}}{\sqrt{4y^3-1} + \sqrt{3}},$$

and then

$$m'^2 = 2 \quad \text{and} \quad m'^2 y = 1 + x',$$

finds first

$$R = m\sqrt{3} \int \frac{dx'}{\sqrt{x'^3 + 3x'^2 + 3}},$$

and then writing $r = \sqrt[6]{3}$, $x = \tan \frac{1}{2}\phi$, and $c^2 = \frac{1}{4}(2 - r^2)$, finds

$$R = \tfrac{1}{2} mr \int \frac{d\phi}{\sqrt{1 - c^2 \sin^2 \phi}};$$

we have therefore only to write $\sin\phi = \operatorname{sn} u$, to modulus

$$c, \ = \ \tfrac{1}{2}\sqrt{2 - \sqrt[3]{3}},$$

and we thence obtain an expression for z in terms of the elliptic functions $\operatorname{sn} u$, $\operatorname{cn} u$, $\operatorname{dn} u$.

Writing x instead of z, and k for c, then

$$m = \sqrt[3]{2}, \quad r = \sqrt[6]{3}; \quad k = \tfrac{1}{2}\sqrt{2 - r^2}, \quad k' = \tfrac{1}{2}\sqrt{2 + r^2}.$$

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Working out the substitutions, the resulting formulæ are

$$x = \frac{2r \operatorname{sn} u \operatorname{dn} u - (1 + \operatorname{cn} u)^2}{2r \operatorname{sn} u \operatorname{dn} u + (1 + \operatorname{cn} u)^2},$$

$$y = \frac{m(1 + \operatorname{cn} u)\,[1 + r^2 + (1 - r^2) \operatorname{cn} u]}{2r \operatorname{sn} u \operatorname{dn} u + (1 + \operatorname{cn} u)^2},$$

where the modulus is k as above; and these values give

$$x^3 + y^3 = 1, \qquad \frac{dx}{(1 - x^3)^{\frac{2}{3}}}, = \frac{-\,dy}{(1 - y^3)^{\frac{2}{3}}} = \tfrac{1}{2} mr\, du.$$

The verification is interesting enough; starting from the expression for x, and for shortness representing it by

$$x = \frac{A - B}{A + B},$$

we have

$$1 - x^3 = \frac{2B(3A^2 + B^2)}{(A + B)^3}, \quad = \frac{m^3(1 + \operatorname{cn} u)^3(3A^2 + B^2)}{[2r \operatorname{sn} u \operatorname{dn} u + (1 + \operatorname{cn} u)^2]^3}.$$

We find

$$3A^2 + B^2 = 12r^2 \operatorname{cn}^2 u \ \operatorname{dn}^2 u + (1 + \operatorname{cn} u)^4,$$

$$= (1 + \operatorname{cn} u)\,[12r^2(1 - \operatorname{cn} u)(k'^2 + k^2 \operatorname{cn}^2 u) + (1 + \operatorname{cn} u)^3],$$

where the term in { } is a perfect cube

$$= [1 + \operatorname{cn} u + r^2(1 - \operatorname{cn} u)]^3.$$

The last-mentioned expression is, in fact,

$$(1 + \operatorname{cn} u)^3 + r^2(1 - \operatorname{cn} u)\,[3(1 + \operatorname{cn} u)^2 + 3r^2(1 - \operatorname{cn} u)(1 + \operatorname{cn} u) + r^4(1 - \operatorname{cn} u)^2],$$

where the second term is

$$= 12r^2(1 - \operatorname{cn} u)\,[\tfrac{1}{4}(1 + \operatorname{cn} u)^2 + \tfrac{1}{4} r^2(1 - \operatorname{cn}^2 u)],$$

that is, it is

$$= 12r^2(1 - \operatorname{cn} u)\,(k'^2 + k^2 \operatorname{cn}^2 u).$$

We have consequently

$$1 - x^3 = \frac{m^3(1 + \operatorname{cn} u)^3\,[1 + r^2 + (1 - r^2) \operatorname{cn} u]^3}{[2r \operatorname{sn} u \operatorname{dn} u + (1 + \operatorname{cn} u)^2]^3},$$

or extracting the cube root $y, = \sqrt[3]{1 - x^3}$, has its foregoing value: and the differential expressions are then verified.

Suppose $y = 1$, we have

$$(m - 1)(1 + \operatorname{cn} u)^2 + mr^2(1 - \operatorname{cn}^2 u) = 2r \operatorname{sn} u \ \operatorname{dn} u,$$

that is,

$$\begin{aligned}(m-1)^2(1+\operatorname{cn}u)^4 &+ 2m(m-1)r^2(1+\operatorname{cn}u)^2(1-\operatorname{cn}u)\\ &+ 3m^2(1+\operatorname{cn}u)(1-\operatorname{cn}u)^2 = r^2(1-\operatorname{cn}u)\{4-4k^2(1-\operatorname{cn}^2u)\},\end{aligned}$$

or observing that the right-hand side is

$$= r^2(1-\operatorname{cn}u)\{(1+\operatorname{cn}u)^2+(1-\operatorname{cn}u)^2+r^2(1+\operatorname{cn}u)(1-\operatorname{cn}u)\},$$

and multiplying by $\frac{1}{1+\operatorname{cn}u}$, the equation becomes

$$\begin{aligned}0 = {}&\tfrac{1}{2}(m-1)^2r^2(1+\operatorname{cn}u)^3+(2m^2-2m+1)(1+\operatorname{cn}u)^2(1-\operatorname{cn}u)\\ &+(m^2-1)r^2(1+\operatorname{cn}u)(1-\operatorname{cn}u)^2-(1-\operatorname{cn}u)^3;\end{aligned}$$

viz. this is

$$0 = \tfrac{1}{2}r^2(m^2-1)(1+\operatorname{cn}u)-(1-\operatorname{cn}u)^3,$$

as is immediately verified: hence writing $\frac{1}{2}r^2 = \dfrac{1}{m^2}$, we have for the value in question, $y=1$,

$$(m^2-1)(1+\operatorname{cn}u)-r^2(1-\operatorname{cn}u)=0,$$

or say

$$m^2(1+\operatorname{cn}u)=(1+\operatorname{cn}u)+r^2(1-\operatorname{cn}u),$$

that is,

$$\operatorname{cn}u=\frac{r^2+1-m^2}{r^2-1+m^2},$$

which is one of the values of $\operatorname{cn}u$ derived from the equation $x=0$; but this equation $x=0$ gives, not the foregoing equation, but

$$m^2(1+\operatorname{cn}u)^3=[(1+\operatorname{cn}u)+r^2(1-\operatorname{cn}u)]^3,$$

viz. the three values of $\operatorname{cn}u$ are the foregoing value and the two values obtained therefrom by changing m into ωm and $\omega^2 m$ respectively, ω being an imaginary cube root of unity. In fact, the curve $x^3+y^3-1=0$ has at the point $x=0$, $y=1$ an inflexion, the tangent being $y=1$, so that this line meets the curve in the point counting three times; but the line $x=0$ meets the curve in the point, and besides in two imaginary points.

805.

NOTE ON ABEL'S THEOREM.

[From the *Proceedings of the Cambridge Philosophical Society*, vol. IV. (1883), pp. 119–122.]

CONSIDERING Abel's theorem in so far as it relates to the first kind of integrals, and as a differential instead of an integral theorem, the theorem may be stated as follows:

We have a fixed curve $f(x,\ y,\ 1) = 0$ of the order m; this implies a relation $f'(x)\,dx + f'(y)\,dy = 0$, between the differentials dx, dy of the coordinates of a point on the curve; and we may therefore write

$$d\omega = \frac{dx}{f'(y)} = -\frac{dy}{f'(x)},$$

and, instead of dx or dy, use $d\omega$ to denote the displacement of a point $(x,\ y)$ on the curve.

Taking for greater simplicity the fixed curve to be a curve without nodes or cusps, and therefore of the deficiency $\frac{1}{2}(m-1)(m-2)$, we consider its intersections by a variable curve $\phi(x,\ y,\ 1) = 0$ of the order n. And then, if $(x,\ y,\ 1)^{m-3}$ denote an arbitrary rational and integral function of $(x,\ y)$ of the order $m-3$, the theorem is that we have between the displacements $d\omega_1$, $d\omega_2$, $\ldots$, $d\omega_{mn}$ of the mn points of intersection, the relation

$$\Sigma(x,\ y,\ 1)^{m-3}\,d\omega = 0,$$

where the left-hand side is the sum of the values of $(x,\ y,\ 1)^{m-3}\,d\omega$ belonging to the mn points of intersection respectively.

For the proof, observe that, varying in any manner the curve ϕ, we obtain

$$\frac{d\phi}{dx}\,dx + \frac{d\phi}{dy}\,dy + \delta\phi = 0,$$

where $\delta\phi$ is that part which depends on the variation of the coefficients, of the whole variation of ϕ; viz. if $\phi = ax^n + bx^{n-1}y + \ldots$, then $\delta\phi = x^n\, da + x^{n-1}y\, db + \ldots$; $\delta\phi$ is thus, in regard to the coordinates $(x,\ y)$, a rational and integral function of the order n.

Writing in this equation

$$dx,\ dy = \frac{df}{dy}\, d\omega,\ -\frac{df}{dx}\, d\omega,$$

the equation becomes

$$\left(\frac{d\phi}{dx}\frac{df}{dy} - \frac{d\phi}{dy}\frac{df}{dx}\right) d\omega + \delta\phi = 0,$$

or say

$$-J(f,\ \phi)\, d\omega + \delta\phi = 0,$$

that is,

$$d\omega = \frac{\delta\phi}{J(f,\ \phi)};$$

and then multiplying each side by the arbitrary function $(x,\ y,\ 1)^{m-3}$, we have

$$\Sigma(x,\ y,\ 1)^{m-3}\, d\omega = \Sigma \frac{(x,\ y,\ 1)^{m-3}}{J(f,\ \phi)}\, \delta\phi,$$

where $\delta\phi$ being of the order n in the variables, the numerator is a rational and integral function of $(x,\ y)$ of the order $m+n-3$: hence by a theorem contained in Jacobi's paper "Theoremata nova algebraica circa systema duarum æquationum inter duas variables," *Crelle*, t. xiv. (1835), pp. 281–288, [*Ges. Werke*, t. iii., pp. 285–294], the sum on the right-hand side is $= 0$: hence the required result $\Sigma(x,\ y,\ 1)^{m-3}\, d\omega = 0$.

Observing that $(x,\ y,\ 1)^{m-3}$ is an arbitrary function, the equation just obtained breaks up into the equations

$$\Sigma\, d\omega = 0,\quad \Sigma x\, d\omega = 0,\quad \Sigma y\, d\omega = 0,\ \ldots,\quad \Sigma x^{m-3}\, d\omega = 0,\ \ldots,\quad \Sigma y^{m-3}\, d\omega = 0,$$

viz. the number of equations is

$$1 + 2 + \ldots + (m-1),\ = \tfrac{1}{2}(m-1)(m-2),$$

which is $= p$, the deficiency of the curve.

Suppose the fixed curve $f(x,\ y,\ 1) = 0$ is a cubic, $m = 3$, and we have the single relation $\Sigma\, d\omega = 0$, where the summation refers to the $3n$ points of intersection of the cubic and of the variable curve of the order n, $\phi(x,\ y,\ 1) = 0$.

In particular, if this curve be a line, $n = 1$, and the equation is $d\omega_1 + d\omega_2 + d\omega_3 = 0$; here the two points $(x_1,\ y_1)$, $(x_2,\ y_2)$ taken at pleasure on the cubic, determine the line, and they consequently determine uniquely the third point of intersection $(x_3,\ y_3)$; there should thus be a single equation giving the displacement $d\omega_3$ in terms of the displacements $d\omega_1$, $d\omega_2$; viz. this is the equation just found

$$d\omega_1 + d\omega_2 + d\omega_3 = 0.$$

So if the variable curve be a conic, $n = 2$; and we have between the displacements of the six points the relation

$$d\omega_1 + d\omega_2 + \ldots + d\omega_6 = 0 :$$

here five of the points determine the conic, and they therefore determine uniquely the sixth point; and there should be between the displacements a single relation as just found.

If the variable curve be a cubic, $n = 3$, and we have between the displacements of the nine points the relation

$$d\omega_1 + d\omega_2 + \ldots + d\omega_9 = 0 :$$

here eight of the points do *not* determine the cubic ϕ, but they nevertheless determine the ninth point, viz. (reproducing the reasoning which establishes this well-known and fundamental theorem as to cubic curves) if $\phi_0 = 0$ be a particular cubic through the 8 points, then the general cubic is $\phi_0 + kf = 0$, and the intersections with $f = 0$ are given by the equations $\phi_0 = 0$, $f = 0$; whence the ninth point is independent of k, and is determined uniquely by the 8 points. There should thus be a single relation between the displacements, viz. this is the relation just found.

And so if the variable curve be a quartic, or curve of any higher order, it appears in like manner that there should be a single relation between the displacements; this relation being in fact the foregoing relation $\Sigma\, d\omega = 0$.

But take the fixed curve to be a quartic, $m = 4$: then we have between the displacements $d\omega$ the relation

$$\Sigma(x,\ y,\ 1)\, d\omega = 0,$$

that is, the three equations

$$\Sigma x\, d\omega = 0, \quad \Sigma y\, d\omega = 0, \quad \Sigma\, d\omega = 0.$$

If the variable curve is a conic, $n = 2$, then there are 8 points of intersection; 5 of these taken at pleasure determine the conic, and they consequently determine the remaining 3 points of intersection: hence there should be 3 equations. And so if the variable curve be a curve of any higher order, then by considerations similar to those made use of in the case where the first curve is a cubic it appears that the number of equations between the displacements $d\omega$ should always be $= 3$.

But if the variable curve be a line, $n = 1$, then the number of the points of intersection is $= 4$: 2 of these taken at pleasure determine the line, and they consequently determine the remaining 2 points of intersection; and the number of equations between the displacements $d\omega$ should thus be $= 2$. But by what precedes, we have the 3 equations

$$d\omega_1 + d\omega_2 + d\omega_3 + d\omega_4 = 0,$$

$$x_1 d\omega_1 + x_2 d\omega_2 + x_3 d\omega_3 + x_4 d\omega_4 = 0,$$

$$y_1 d\omega_1 + y_2 d\omega_2 + y_3 d\omega_3 + y_4 d\omega_4 = 0;$$

here the 4 points of intersection are on a line $y = ax + b$; we have therefore $y_1 = ax_1 + b, \ldots,$ $y_4 = ax_4 + b$; the equations between the $d\omega$'s give

$$(y_1 - ax_1 - b)\, d\omega_1 + \ldots + (y_4 - ax_4 - b)\, d\omega_4 = 0,$$

that is, is a single relation $0 = 0$; or the 3 equations thus reduce themselves to 2 independent equations.

Again, if the fixed curve be a quintic, $m = 5$, there are here between the displacements the 6 equations

$$\Sigma x^2\, d\omega = 0, \quad \Sigma xy\, d\omega = 0, \quad \Sigma y^2\, d\omega = 0,$$

$$\Sigma x\, d\omega = 0, \quad \Sigma y\, d\omega = 0, \quad \Sigma\, d\omega = 0;$$

the two cases in which the number of independent equations is less than 6 are (i) when the variable curve is a line, and (ii) when the variable curve is a conic. For the line $n = 1$, and the number should be $= 3$. We have the above 6 equations; but the equation of the line is $ax + by + c = 0$, that is, we have $ax_1 + by_1 + c = 0$, &c.; we deduce the 3 identical equations

$$\Sigma x(ax + by + c) = 0, \quad \Sigma y(ax + by + c) = 0, \quad \Sigma(ax + by + c) = 0,$$

and the number of independent equations is thus $6 - 3, = 3$ as it should be.

So when the variable curve is a conic, $n = 2$; the number of independent equations should be $= 5$. The points of intersection lie on a conic $(a,\ b,\ c,\ f,\ g,\ h\langle x,\ y,\ 1\rangle)^2 = 0$; we have therefore the several equations $(a,\ b,\ c,\ f,\ g,\ h\langle x_1,\ y_1,\ 1\rangle)^2 = 0$, &c.: we have therefore the single identical equation

$$\Sigma(a,\ b,\ c,\ f,\ g,\ h\langle x,\ y,\ 1\rangle)^2\, d\omega = 0,$$

and the number of independent equations is $6 - 1, = 5$ as it should be.

Obviously the like considerations apply to the case where the fixed curve is a curve of any given order whatever.

806.

DETERMINATION OF THE ORDER OF A SURFACE.

[From the *Messenger of Mathematics*, vol. xii. (1883), pp. 29–32.]

[On p. lxx of the Prolegomena to C. Taylor's *Introduction to the Geometry of Conics* (1881) occurs the following passage:

"*Proof and extension of Newton's Descriptio Organica.*

"Let two angles AOB and $A\omega B$ of given magnitudes turn about O and ω respectively, and let the intersection A trace a curve of the nth order. For a given position of the arm OB there are n positions of A and therefore n of B. When OB is in the position $O\omega$ the n B's coincide with ω, which is therefore an n-fold point on the locus of B, as is also the point O; and since any line through O (or ω) meets the locus of B in n other points, the locus is of the order $2n$. Its order is the same when $A\omega B$ is a zero-angle or straight line.

"Let a given trihedral angle $O(ABC)$—or a plane OBC and a line OA rigidly attached to it—turn about O, and let a variable plane through a fixed point ω meet OA in A and the plane OBC in BC; then if the line BC describes a ruled surface of the order n the point A describes a surface of the order $4n$."

And in a foot-note it is stated that the author is indebted to Professor Cayley for the determination of the order of this surface. The following paper contains Professor Cayley's determination, which was communicated by him to Mr Taylor.]*

Lemma. Take $(a,\ b,\ c,\ f,\ g,\ h)$ the six coordinates of a line; these are connected by the equation

$$af + bg + ch = 0.$$

[* l.c., p. 29.]

In order that the line may belong to a ruled surface, we must have between the coordinates three more equations, say these are

$$F\,(a,\ b,\ c,\ f,\ g,\ h) = 0,$$
$$G\,(\quad ” \quad) = 0,$$
$$H\,(\quad ” \quad) = 0,$$

of the orders p, q, r respectively; then the scroll is of the order $n = 2pqr$.

For expressing that the line meets an arbitrary line $(a',\ b',\ c',\ f',\ g',\ h')$, we have the linear relation

$$f'a + g'b + h'c + a'f + b'g + c'h = 0;$$

the five relations determine the ratios $a : b : c : f : g : h$, and the number of systems of values is = product of orders $= 2 \cdot p \cdot q \cdot r \cdot 1,\ = 2pqr$, viz. this is the number of lines meeting the arbitrary line; or, what is the same thing, it is = order of ruled surface.

Consider now a trihedral angle $OABC$ rotating about a fixed point O which may be taken for the origin, and consider a fixed point ω. Let the lines OB, OC each meet a line L, and the plane ωL intersect OA in a point P; then it is to be shown that, if L is a line of a ruled surface of the order n, the locus of P is a surface of the order $4n$.

Observe that, for a given position of the line L, the position of the lines OB, OC is not determinate, but that the angle BOC has any position at pleasure in the plane OL, or say it rotates round the line ON which is the normal at O to the plane OL; the line OA therefore also rotates about ON, being always inclined to it at a determinate angle θ, or the locus of OA is a right cone axis ON and semi-aperture $= \theta$.

The points P are the intersections of the several lines of the cone with the fixed plane ωL, or say that (for the given position of L) the locus of P is the conic C which is the intersection of the cone in question by the plane ωL. And then varying the position of L, the required surface is the locus of the corresponding conics C.

Take now

$$Ax + By + Cz + D = 0, \quad A'x + B'y + C'z + D' = 0,$$

for the equations of a particular line L; then writing

$$AD' - A'D,\ BD' - B'D,\ CD' - C'D,\ BC' - B'C,\ CA' - C'A,\ AB' - A'B = (a,\ b,\ c,\ f,\ g,\ h)$$

respectively, these are the six coordinates of the line L.

The equation of the plane OL is

$$(AD' - A'D)\,x + (BD' - B'D)\,y + (CD' - C'D)\,z = 0,$$

that is,

$$ax + by + cz = 0; 6 - 2$$

and the equations of the normal ON are therefore

$$\frac{x}{a} = \frac{y}{b} = \frac{z}{c};$$

we have therefore for the cone

$$\frac{ax + by + cz}{\sqrt{(a^2 + b^2 + c^2)}\,\sqrt{(x^2 + y^2 + z^2)}} = \cos\theta;$$

or if, for convenience, $\cos^2\theta = k$, then the equation of the cone is

$$(ax + by + cz)^2 = k\,(a^2 + b^2 + c^2)\,(x^2 + y^2 + z^2);$$

say this is

$$[a^2 - k\,(a^2 + b^2 + c^2),\ b^2 - k\,(a^2 + b^2 + c^2),\ c^2 - k\,(a^2 + b^2 + c^2),\ bc,\ ca,\ ab\langle x,\ y,\ z\rangle]^2 = 0. \quad \ldots(1)$$

Taking then $(x_0,\ y_0,\ z_0)$ for the coordinates of the point ω, the equation of the plane ωL is

$$\frac{Ax + By + Cz + D}{Ax_0 + By_0 + Cz_0 + D} = \frac{A'x + B'y + C'z + D'}{A'x_0 + B'y_0 + C'z_0 + D'},$$

viz. it is

$$(BC' - B'C)(yz_0 - y_0z) + \ldots + (AD' - A'D)(x - x_0) + \ldots = 0,$$

that is,

$$f(yz_0 - y_0z) + g(zx_0 - z_0x) + h(xy_0 - x_0y) + a(x - x_0) + b(y - y_0) + c(z - z_0) = 0,$$

or say

$$x(hy_0 - gz_0 + a) + y(-hx_0 + fz_0 + b) + z(gx_0 - fy_0 + c) + (-ax_0 - by_0 - cz_0) = 0, \quad \ldots(2)$$

viz. (1) and (2) are the equations of the conic C, and the coordinates $(a,\ b,\ c,\ f,\ g,\ h)$ satisfy of course the equation

$$af + bg + ch = 0. \qquad \ldots\ldots(3)$$

Considering now the line L as belonging to a ruled surface, the coordinates $(a,\ \ldots)$ satisfy as before three equations

$$\begin{aligned} F\,(a,\ b,\ c,\ f,\ g,\ h) &= 0, \qquad \ldots\ldots(4)\\ G\,(\quad ” \quad) &= 0, \qquad \ldots\ldots(5)\\ H\,(\quad ” \quad) &= 0, \qquad \ldots\ldots(6)\end{aligned}$$

of the orders $p,\ q,\ r$ respectively, and we can from the six equations eliminate $a,\ b,\ c,\ f,\ g,\ h$. The resulting equation $\nabla = 0$ contains the coefficients of (1) in the order $1 \cdot 2 \cdot p \cdot q \cdot r = 2pqr$ (which is the product of the orders of the other 5 equations), and the coefficients of (2) in the order $2 \cdot 2 \cdot p \cdot q \cdot r$ (which is the product of the orders of the other 5 equations). But the coefficients of (1) being quadric functions of $(x,\ y,\ z)$, and those of (2) being linear functions of $(x,\ y,\ z)$, the aggregate order in $(x,\ y,\ z)$ is

$$2 \cdot 2pqr + 1 \cdot 4pqr,\ = 8pqr;$$

or, since the order of the ruled surface is $n,\ = 2pqr$, the order of the locus is $4n$; which is the above-stated theorem.

807.

A PROOF OF WILSON'S THEOREM.

[From the *Messenger of Mathematics*, vol. xii. (1883), p. 41.]

LET n be a prime number; and imagine n points, the vertices of a regular polygon; any polygon which can be formed with these n points as vertices is either regular or else it is one of a set of n equal and similar polygons. For instance, $n = 5$, the polygon as shown in the figure is one of a set of 5 equal and similar

[Figure: 5-pointed star polygon labelled 1 2 3 4 5 *with sides* 13254*]*

polygons; in fact, if the points taken in their cyclical order, but beginning at pleasure with any one of the 5 points are called 1, 2, 3, 4, 5, then we have 5 such polygons 13254; and so in general. The whole number of polygons is $\frac{1}{2} \cdot 1 \cdot 2 \cdot 3 \ldots (n-1)$; and the number of the regular polygons is $\frac{1}{2}(n-1)$; hence the number of the remaining polygons is $= \frac{1}{2}(n-1)\,[1 \cdot 2 \ldots (n-2) - 1]$; and this number must therefore be divisible by n; that is, $1 \cdot 2 \ldots (n-1) - n + 1$ is divisible by n; or, what is the same thing, $1 \cdot 2 \ldots (n-1) + 1$ is divisible by n, which is the theorem in question.

808.

NOTE ON A FORM OF THE MODULAR EQUATION IN THE TRANSFORMATION OF THE THIRD ORDER.

[From the *Messenger of Mathematics*, vol. xii. (1883), pp. 173, 174.]

In my *Treatise on Elliptic Functions*, pp. 214–216, writing only $\frac{1}{J}$, $\frac{1}{J'}$ instead of Ω, Ω', and α, β instead of α', β', I have shown as follows: viz. if k, λ denote as usual the original modulus, and the transformed modulus, and if

$$J = \frac{(k^4 + 14k^2 + 1)^3}{108k^4(1-k^2)^4}, \quad J' = \frac{(\lambda^4 + 14\lambda^2 + 1)^3}{108\lambda^4(1-\lambda^2)^4},$$

then the relation between J and J' can be found by the elimination of α, β from the equations

$$\alpha + \beta = 1,$$

$$J = \frac{(1+8\alpha)^3}{64\alpha(1-\alpha)^3}, \quad J' = \frac{(1+8\beta)^3}{64\beta(1-\beta)^3}.$$

By a very slight change we obtain the result given by Prof. Klein in his paper, "Ueber die Transformation der elliptischen Functionen, &c.," *Math. Ann.* t. xiv. (1879), pp. 111–172; viz., see p. 143, the relation is to be obtained by the elimination of τ, τ' from the equation $\tau\tau' = 1$, and the equations

$$J : J-1 : 1 = (\tau-1)(9\tau-1)^3 : (27\tau^2 - 18\tau - 1)^2 : -64\tau;$$

$$J' : J'-1 : 1 = (\tau'-1)(9\tau'-1)^3 : (27\tau'^2 - 18\tau' - 1)^2 : -64\tau';$$

these last equations being equivalent to two equations only in virtue of the identity

$$(\tau-1)(9\tau-1)^3 + 64\tau = (27\tau^2 - 18\tau - 1)^2,$$

and the like identity in τ'.

In fact, writing $\alpha = \frac{\tau}{\tau-1}$, $\beta = \frac{\tau'}{\tau'-1}$, the equation $\alpha + \beta = 1$ becomes $\tau\tau' = 1$; and then for α, β substituting their values, we have

$$J = \frac{(9\tau-1)^3(\tau-1)}{-64\tau}, \quad J' = \frac{(9\tau'-1)^3(\tau'-1)}{-64\tau'},$$

which are the formulæ in question.

809.

SCHRÖTER'S CONSTRUCTION OF THE REGULAR PENTAGON.

[From the *Messenger of Mathematics*, vol. xii. (1883), p. 177.]

THE following construction of the regular pentagon, analogous to the more complicated one for the polygon of 17 sides, is given in the paper, H. Schröter, Zur v. Staudtschen "Construction des regulären Siebenzehnecks," *Crelle*, t. lxxv. (1873), pp. 13–24.

Take in a circle AB, CD diameters at right angles to each other, and at C, D draw in the directions A to B and B to A respectively, the lines Cc, $=4$ radii, and Dd, $=1$ radius; draw cd meeting the circle in the points E and F; draw CE and CF cutting AB in the points e and f respectively, and through these at right angles to AB draw the chords 34 and 25 respectively, then we have $A2345$ a regular pentagon.

810.

NOTE ON A SYSTEM OF EQUATIONS.

[From the *Messenger of Mathematics*, vol. xii. (1883), pp. 191, 192.]

THE equations are

$$x^2 = ax + by, \quad xy = cx + dy, \quad y^2 = ex + fy,$$

where

$$\frac{b}{d} = \frac{a-d}{c-f} = \frac{c}{e};$$

or, what is the same thing, if a, b, c, d are given, then

$$e = \frac{cd}{b}, \quad f = c - \frac{d\,(a-d)}{b};$$

and this being so, the equations are equivalent to two independent equations; viz. starting from the first and the second equations, we have

$$dx^2 - bxy = (ad - bc)\,x,$$

that is,

$$dx - by = (ad - bc);$$

and thence

$$dxy - by^2 = (ad - bc)\,y,$$

or

$$d\,(ex + dy) - by^2 = (ad - bc)\,y;$$

which, attending to the values of e and f, is the third equation

$$y^2 = ex + fy.$$

We have

$$\frac{x}{y} = \frac{ax + by}{cx + dy}, \quad \frac{y}{x} = \frac{ex + fy}{cx + dy};$$

that is,

$$cx^2 - (a-d)\,xy - by^2 = 0,$$
$$ex^2 - (c-f)\,xy - dy^2 = 0,$$

and eliminating the $(x,\ y)$ from these equations we have an equation $\Omega = 0$ as the condition that the original three equations may have a single common root; the before-mentioned equations $\dfrac{b}{d} = \dfrac{a-d}{c-f} = \dfrac{c}{e}$ are the conditions in order that the three equations may have two common roots, that is, that there may be two systems of $(x,\ y)$ satisfying the three equations.

We have, moreover, $y\,(x-d) = ex$, $x\,(y-c) = dy$, and substituting these values, say $y = \dfrac{ex}{x-d}$ and $x = \dfrac{dy}{y-c}$, in the first and third equations respectively, they become

$$x - a = \frac{bc}{x-d}, \quad y - f = \frac{de}{y-c},$$

that is,

$$x^2 - (a+d)\,x + ad - bc = 0,$$
$$y^2 - (c+f)\,y + cf - de = 0,$$

which are quadric equations for x and y respectively; it is easy to express the second equation (like the first) in terms of $(a,\ b,\ c,\ d)$, and the first equation (like the second) in terms of $(c,\ d,\ e,\ f)$, but the forms are less simple.

Suppose $(a,\ b,\ c,\ d) = (-1,\ -1,\ 1,\ 0)$, then we have $(e,\ f) = (0,\ 1)$, the two equations in $x : y$ become $x^2 + xy + y^2 = 0$, and $0 = 0$ respectively; those in x and y become $x^2 + x + 1 = 0$, $y^2 - 2y + 1 = 0$ respectively; this is right, for the three equations are

$$x^2 = -x - y, \quad xy = x, \quad y^2 = y;$$

viz. from the third equation we have $y = 1$, a value satisfying the second equation, and then the first equation becomes $x^2 + x + 1 = 0$, or, if we please, $x^2 + xy + y^2 = 0$; the values in fact being $x = \omega$, an imaginary cube root of unity, and $y = 1$.

In the general case, the values $(x,\ y)$ may be regarded as units in a complex numerical theory, viz. if $(a,\ b,\ c,\ d,\ e,\ f)$ are integers, and $p,\ q,\ p',\ q',\ P,\ Q$ are also integers, then the product of the two complex integers $px + qy$ and $p'x + q'y$ will be a complex integer $Px + Qy$.

811.

ON THE LINEAR TRANSFORMATION OF THE THETA FUNCTIONS.

[From the *Messenger of Mathematics*, vol. xiii. (1884), pp. 54–60.]

THE functions referred to are the single Theta Functions; these may be defined as doubly infinite products, as was in fact done in my "Mémoire sur les fonctions doublement périodiques," *Liouv.* t. x. (1845), pp. 385–420, [25]; and it is interesting to consider from this point of view the theory of their linear transformation: this I propose to do in the present paper, adopting throughout the notation of Smith's* "Memoir on the Theta and Omega Functions."

The periods K, iK' are, in general, imaginary quantities

$$K = A + Bi,$$

$$iK' = C + Di,$$

where $AD - BC$ is positive; writing then $\omega = \dfrac{iK'}{K}$, and $q = e^{i\pi\omega}$, also for shortness

$$(q1) = 2q^{\frac{1}{8}} \prod_1^\infty (1 - q^{2n})^3,$$

where $q^{\frac{1}{8}}$ denotes $e^{\frac{i\pi\omega}{8}}$, the expression of the odd theta-function $\vartheta_1(x,\ \omega)$ as a doubly infinite product is

$$\vartheta_1(x,\ \omega) = (q1)\, x \prod{}' \left(1 + \frac{x}{m\pi + n\pi\omega}\right), \quad \left(\frac{\mu}{\nu} = \infty\right),$$

where $(m,\ n)$ have any positive or negative integer values (the combination $m = 0,\ n = 0$ excluded) from $m = -\mu$ to μ, and $n = -\nu$ to ν, μ and ν being each ultimately infinite but so that μ is infinite in comparison with ν; this condition in regard to the limits is indicated by $\mu/\nu = \infty$; and similarly $\nu/\mu = \infty$ would indicate that ν was infinite in comparison with μ.

[* Smith's *Collected Mathematical Papers*, vol. ii., pp. 415–621.]

The condition as to the limits might be that $(m,\ n)$ have any positive or negative values (excluding as before) such that the modulus of $m+n\omega$ does not exceed a positive value T, which is ultimately taken to be infinite; this condition may be indicated by mod $=\infty$.

The values of the double product corresponding to the different conditions as to the limits are not equal, but they differ only by an exponential factor, the exponent being a multiple of x^2; we thus have

$$x\,\Pi\Pi\left(1+\frac{x}{mK+niK'}\right)\left(\frac{\mu}{\nu}=\infty\right)=\exp\left(\nabla x^2\right)x\,\Pi\Pi\left(1+\frac{x}{mK+niK'}\right)(\mathrm{mod}=\infty),$$

where ∇ is a determinate value, depending on K and K'; and similarly

$$x\,\Pi\Pi\left(1+\frac{x}{m\Lambda+ni\Lambda'}\right)\left(\frac{\mu}{\nu}=\infty\right)=\exp\left(\Box x^2\right)x\,\Pi\Pi\left(1+\frac{x}{m\Lambda+ni\Lambda'}\right)(\mathrm{mod}=\infty),$$

where $\Box$ is a determinate value depending in like manner on Λ, Λ'.

We have, then, as above

$$\begin{aligned}\vartheta_1(x,\ \omega)&=(q1)\,x\,\Pi\Pi\left(1+\frac{x}{m\pi+n\pi\omega}\right)\left(\frac{\mu}{\nu}=\infty\right)\\&=(q1)\frac{\pi}{K}\cdot\frac{Kx}{\pi}\,\Pi\Pi\left(1+\frac{\dfrac{Kx}{\pi}}{mK+niK'}\right)\\&=(q1)\frac{\pi}{K}\,\exp\left(\nabla\frac{K^2x^2}{\pi^2}\right)\frac{Kx}{\pi}\,\Pi\Pi\left(1+\frac{\dfrac{Kx}{\pi}}{mK+niK'}\right)(\mathrm{mod}=\infty),\end{aligned}$$

viz. we have thus defined $\vartheta_1(x,\ \omega)$ as a doubly infinite product with the limiting condition $(\mathrm{mod}=\infty)$; if for x we write $\dfrac{\pi x}{h}$, h arbitrary, we have

$$\vartheta_1\left(\frac{\pi x}{h},\ \omega\right)=(q1)\frac{\pi}{K}\,\exp\left(\nabla\frac{K^2x^2}{h^2}\right)\frac{Kx}{h}\,\Pi\Pi\left(1+\frac{\dfrac{Kx}{h}}{mK+niK'}\right)(\mathrm{mod}=\infty),$$

and similarly, if $\Omega=\dfrac{i\Lambda'}{\Lambda}$, $Q=e^{i\pi\Omega}$, then

$$\begin{aligned}\vartheta_1\left\{(a+b\Omega)\frac{\pi x}{h},\ \Omega\right\}&=(Q1)\frac{\pi}{\Lambda}\,\exp\left\{(a+b\Omega)^2\,\Box\,\Lambda^2\,\frac{x^2}{h^2}\right\}\\&\times(a+b\Omega)\frac{\Lambda x}{h}\,\Pi\Pi\left(1+\frac{(a+b\Omega)\dfrac{\Lambda x}{h}}{m\Lambda+ni\Lambda'}\right)(\mathrm{mod}=\infty).\end{aligned}$$

In the case of a linear transformation, we have

$$\omega=\begin{vmatrix}a, & b\\ c, & d\end{vmatrix}\Theta\,\Omega,\quad\text{that is,}\quad \omega=\frac{c+d\Omega}{a+b\Omega},7-2$$

where a, b, c, d are positive or negative integers such that $ad - bc = +1$; it is to be shown that the two infinite products are in this case identical; this being so, we have

$$\frac{\vartheta_1\left\{(a+b\Omega)\dfrac{\pi x}{h},\ \Omega\right\}}{\vartheta_1\left\{\dfrac{\pi x}{h},\ \omega\right\}} = \frac{(Q1)}{(q1)}\ \exp\left\{(a+b\Omega)^2 \,\square\, \Lambda^2 - \nabla\, K^2\right\} \frac{x^2}{h^2},$$

viz. the two functions differ only by a constant factor and by an exponential factor, the exponent being a multiple of x^2: after all reductions, this factor is found to be

$$= \exp\left(-\,i\pi b\,(a+b\Omega)\,\frac{x^2}{h^2}\right).$$

We have

$$\omega = \frac{c+d\Omega}{a+b\Omega},$$

or since

$$\omega = \frac{iK'}{K}, \quad \Omega = \frac{i\Lambda'}{\Lambda},$$

this is

$$\frac{iK'}{K} = \frac{c\Lambda + di\Lambda'}{a\Lambda + bi\Lambda'},$$

or say

$$\frac{1}{M}K = a\Lambda + bi\Lambda',$$

$$\frac{1}{M}iK' = c\Lambda + di\Lambda',$$

either of which equations may be taken as a definition of the multiplier M. We have

$$\frac{K}{M\Lambda} = a + bi\Omega,$$

$$\begin{aligned}\frac{1}{M}(mK + niK') &= (ma+cn)\,\Lambda + (bm+dn)\,i\Lambda' \\ &= m'\Lambda + n'i\Lambda',\end{aligned}$$

if

$$m' = am + cn,$$

$$n' = bm + dn.$$

Here to any integer values of $(m,\ n)$ there correspond integer values of m', n'; and conversely, in virtue of the equation $ad - bc = 1$, to any integer values of m', n' there correspond integer values of m, n. The two products are

$$\Pi\Pi\left(1 + \frac{\dfrac{Kx}{h}}{m'\Lambda + n'i\Lambda'}\right), \ (\mathrm{mod} = \infty),$$

$$\Pi\Pi\left(1 + \frac{(a+b\Omega)\dfrac{\Lambda x}{h}}{m\Lambda + ni\Lambda'}\right), \ (\mathrm{mod} = \infty).$$

But, as above, we have $\dfrac{K}{M} = (a + b\Omega)\,\Lambda$: and then, observing that in the first of the two products we may for m', n' write m, n, it at once appears that the two products are identical.

The exponential factor, writing therein $(a + b\Omega)\,\Lambda = \dfrac{K}{M}$, becomes

$$\exp\left[\left(\frac{\square}{M^2} - \nabla\right)\frac{K^2x^2}{h^2}\right].$$

The values of ∇, $\square$ are at once obtained by means of a formula* given in my Memoir, viz. we have

$$\nabla = -\tfrac{1}{2}(B + \beta),$$

where

$$B = \frac{\pi(\omega\nu + \omega'\nu')}{\Pi\,\Upsilon\ \mathrm{mod}\ (\omega\nu - \omega'\nu)},$$

$$\beta = \frac{\pi i}{\Pi\,\Upsilon}\frac{(\omega\nu - \omega'\nu')}{\mathrm{mod}(\omega\nu - \omega'\nu)}.$$

Comparing with the present notation

$$\Omega = \omega + \omega' i,\ = A + Bi = K,$$

$$\Upsilon = \nu + \nu' i,\ = C + Di = K'i,$$

so that Ω, Υ denote K, $K'i$, and ω, ω', ν, ν' denote A, B, C, D respectively: $\omega\nu - \omega'\nu$ is thus $= AD - BC$, which has been assumed to be positive; hence also $\mathrm{mod}\,(\omega\nu - \omega'\nu) = AD - BC$, and the formula becomes

$$\nabla = -\tfrac{1}{2}\pi\left\{\frac{AC + BD}{i\,(AD - BC)} + 1\right\}\frac{1}{KK'}.$$

Now writing

$$\Lambda = A_1 + B_1 i,$$

$$i\Lambda' = C_1 + D_1 i,$$

then we have

$$\frac{1}{M}(A + Bi) = a\Lambda + bi\Lambda' = a\,(A_1 + B_1 i) + b\,(C_1 + D_1 i),$$

$$\frac{1}{M}(C + Di) = c\Lambda + di\Lambda' = c\,(A_1 + B_1 i) + d\,(C_1 + D_1 i);$$

consequently, if

$$M = \rho\,(\cos\theta + i\sin\theta),$$

[* *Collected Mathematical Papers*, t. i., p. 164. The denominator factor $i^2\Upsilon$ has been omitted (p. 165) by mistake.]

we have

$$\begin{aligned}
\frac{1}{\rho}(\quad A\cos\theta + B\sin\theta) &= aA_1 + bC_1,\\
\frac{1}{\rho}(-A\sin\theta + B\cos\theta) &= aB_1 + bD_1,\\
\frac{1}{\rho}(\quad C\cos\theta + D\sin\theta) &= cA_1 + dC_1,\\
\frac{1}{\rho}(-C\sin\theta + D\cos\theta) &= cB_1 + dD_1,
\end{aligned}$$

and thence

$$\frac{1}{\rho^2}(AD - BC) = (ad - bc)(A_1D_1 - B_1C_1), \ = (A_1D_1 - B_1C_1).$$

Hence $A_1D_1 - B_1C_1$ is positive, and we have

$$\Box = -\tfrac{1}{2}\pi\left\{\frac{A_1C_1 + B_1D_1}{i\,(A_1D_1 - B_1C_1)} + 1\right\}\frac{1}{\Lambda\Lambda'}.$$

Take K_1 the conjugate of K, Λ_1 the conjugate of Λ, then

$$K_1 = A - Bi, \quad \Lambda_1 = A_1 - B_1 i,$$

$$iK' = C + Di, \quad i\Lambda' = C_1 + D_1 i.$$

We have

$$iK_1K' = AC + BD + i\,(AD - BC),$$

and therefore

$$\frac{K_1K'}{AD - BC} = \frac{AC + BD}{i\,(AD - BC)} + 1, \quad \nabla = \frac{-\frac{1}{2}\pi}{AD - BC}\cdot\frac{K_1}{K},$$

and similarly

$$\Box = \frac{-\frac{1}{2}\pi}{A_1D_1 - B_1C_1}\cdot\frac{\Lambda_1}{\Lambda}.$$

The exponential is

$$\left(\frac{\Box}{M^2} - \nabla\right)\frac{K^2x^2}{h^2},$$

and we have

$$\frac{\Box}{M^2} - \nabla = \frac{-\frac{1}{2}\pi}{M^2(A_1D_1 - B_1C_1)}\,\frac{\Lambda_1}{\Lambda} + \frac{\frac{1}{2}\pi}{AD - BC}\,\frac{K_1}{K},$$

which is

$$\begin{aligned}
&= \frac{-\frac{1}{2}\pi}{M^2(A_1D_1 - B_1C_1)}\,\frac{\Lambda_1}{\Lambda} + \frac{\frac{1}{2}\pi}{\rho^2(A_1D_1 - B_1C_1)}\,\frac{K_1}{K}\\
&= \frac{-\frac{1}{2}\pi}{A_1D_1 - B_1C_1}\left(\frac{1}{M^2}\,\frac{\Lambda_1}{\Lambda} - \frac{1}{\rho^2}\,\frac{K_1}{K}\right).
\end{aligned}$$

But $\rho/M = \cos\theta - i\sin\theta$, or calling this for a moment P, then $1/M^2 = P^2/\rho^2$, and the formula may be written

$$\frac{\Box}{M^2} - \nabla = \frac{-\frac{1}{2}\pi P}{\rho^2(A_1D_1 - B_1C_1)}\left(P\frac{\Lambda_1}{\Lambda} - P^{-1}\frac{K_1}{K}\right)$$

$$= \frac{-\frac{1}{2}\pi P}{\rho^2(A_1D_1 - B_1C_1)}\frac{1}{K\Lambda}\left\{(\cos\theta - i\sin\theta)\,\Lambda_1 K - (\cos\theta + i\sin\theta)\,\Lambda K_1\right\}.$$

The term in { } is

$$\begin{aligned}
&(\cos\theta - i\sin\theta)\,(A + Bi)\,(A_1 - B_1 i) - (\cos\theta + i\sin\theta)\,(A - Bi)\,(A_1 + B_1 i)\\
&\quad = 2\cos\theta\,[-(AB_1 - A_1B)\,i] - 2i\sin\theta\,(AA_1 + BB_1),\\
&\quad = -2i\,\{(AB_1 - A_1B)\,\cos\theta + (AA_1 + BB_1)\,\sin\theta\},\\
&\quad = -2i\,\{B_1\,(A\cos\theta + B\sin\theta) - A_1\,(-A\sin\theta + B\cos\theta)\},\\
&\quad = -2i\rho\,\{B_1\,(aA_1 + bC_1) - A_1\,(aB_1 + bD_1)\},\\
&\quad = +2ib\rho\,(A_1D_1 - B_1C_1).
\end{aligned}$$

Hence

$$\frac{\Box}{M^2} - \nabla = \frac{-\frac{1}{2}\pi P}{\rho^2(A_1D_1 - B_1C_1)}\,2ib\rho\,(A_1D_1 - B_1C_1)\frac{1}{K\Lambda}$$

$$= -\frac{i\pi bP}{\rho}\cdot\frac{1}{K\Lambda} = -\frac{i\pi b}{M}\cdot\frac{1}{K\Lambda},$$

and the exponential thus is

$$= \exp\left(-\frac{i\pi b}{M}\cdot\frac{1}{K\Lambda}\cdot\frac{K^2x^2}{h^2}\right) = \exp\left(-i\pi b\,\frac{K}{M\Lambda}\cdot\frac{x^2}{h^2}\right);$$

or, since $\dfrac{K}{M\Lambda} = (a + b\Omega)$, this is

$$= \exp\left(-i\pi b\,(a + b\Omega)\,\frac{x^2}{h^2}\right);$$

and we have thus the required formula

$$\frac{\vartheta_1\left\{(a + b\Omega)\dfrac{\pi x}{h},\ \Omega\right\}}{\vartheta_1\left\{\dfrac{\pi x}{h},\ \omega\right\}} = \frac{(Q1)}{(q1)}\,(a + b\Omega)\,\exp\left(-i\pi b\,(a + b\Omega)\,\frac{x^2}{h^2}\right).$$